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GitHub Link	https://github.com/chaitramanohar/MLA0402-DEEP-LEARNING

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I T.Chaitra(192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.Chaitra

Name of the student:T.chaitra

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

5. Proposed Methodology

Explain the overall methodology adopted in the article.

Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers → Classification/Decoder → Output

Identify the role of the major components in the architecture.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score, Loss, Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting

Limited generalization, Model complexity.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

ARTICLE REVIEW — TRANSFER LEARNING

1. Article Details

Title: *Toward accurate Alzheimer's detection: transfer learning with ResNet50 for MRI-based diagnosis*

Authors: Jabli Mohamed Amine and Mourad M.

Journal: *Frontiers in Neuroscience*

Year: 2025

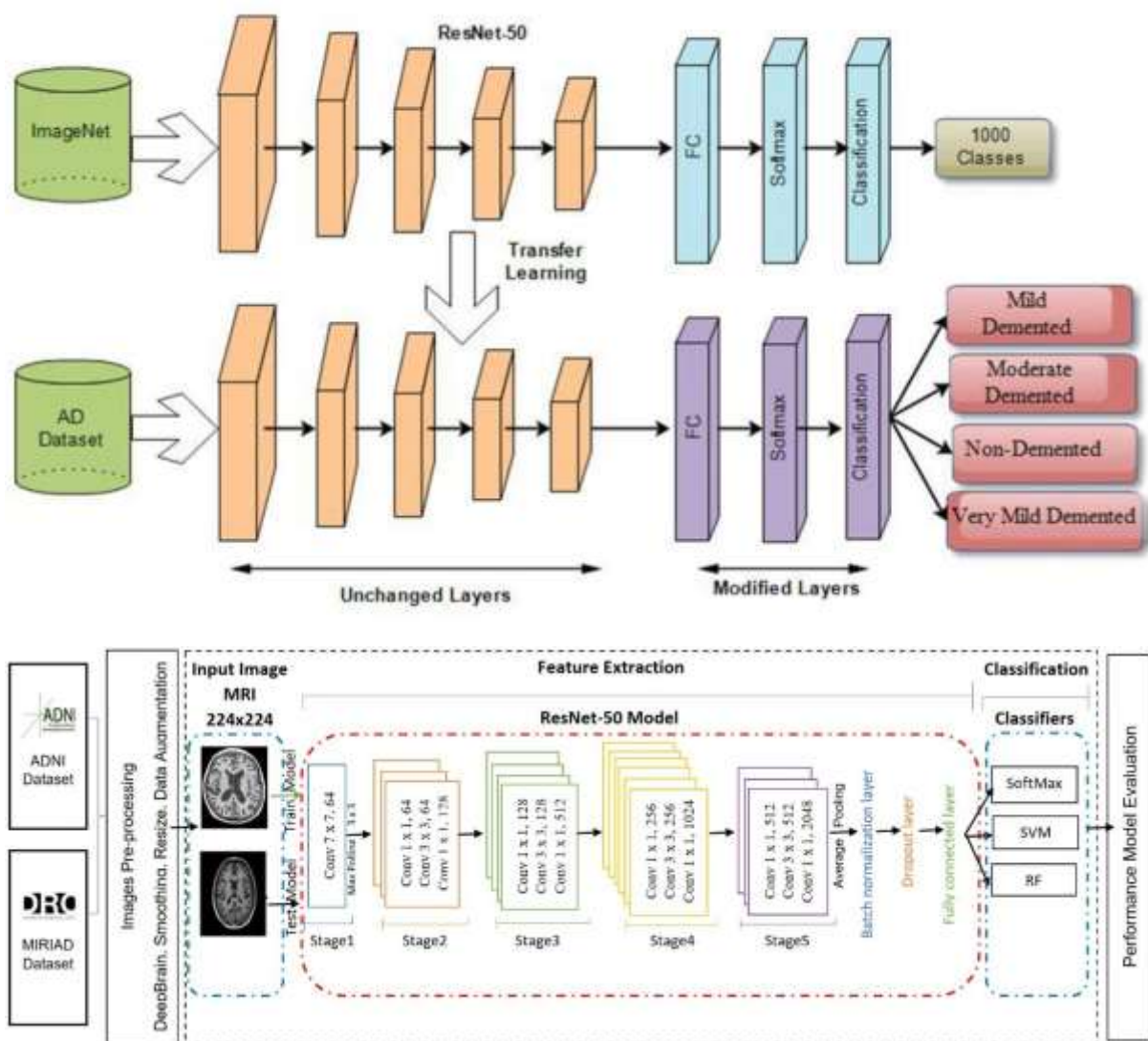
Volume: 19

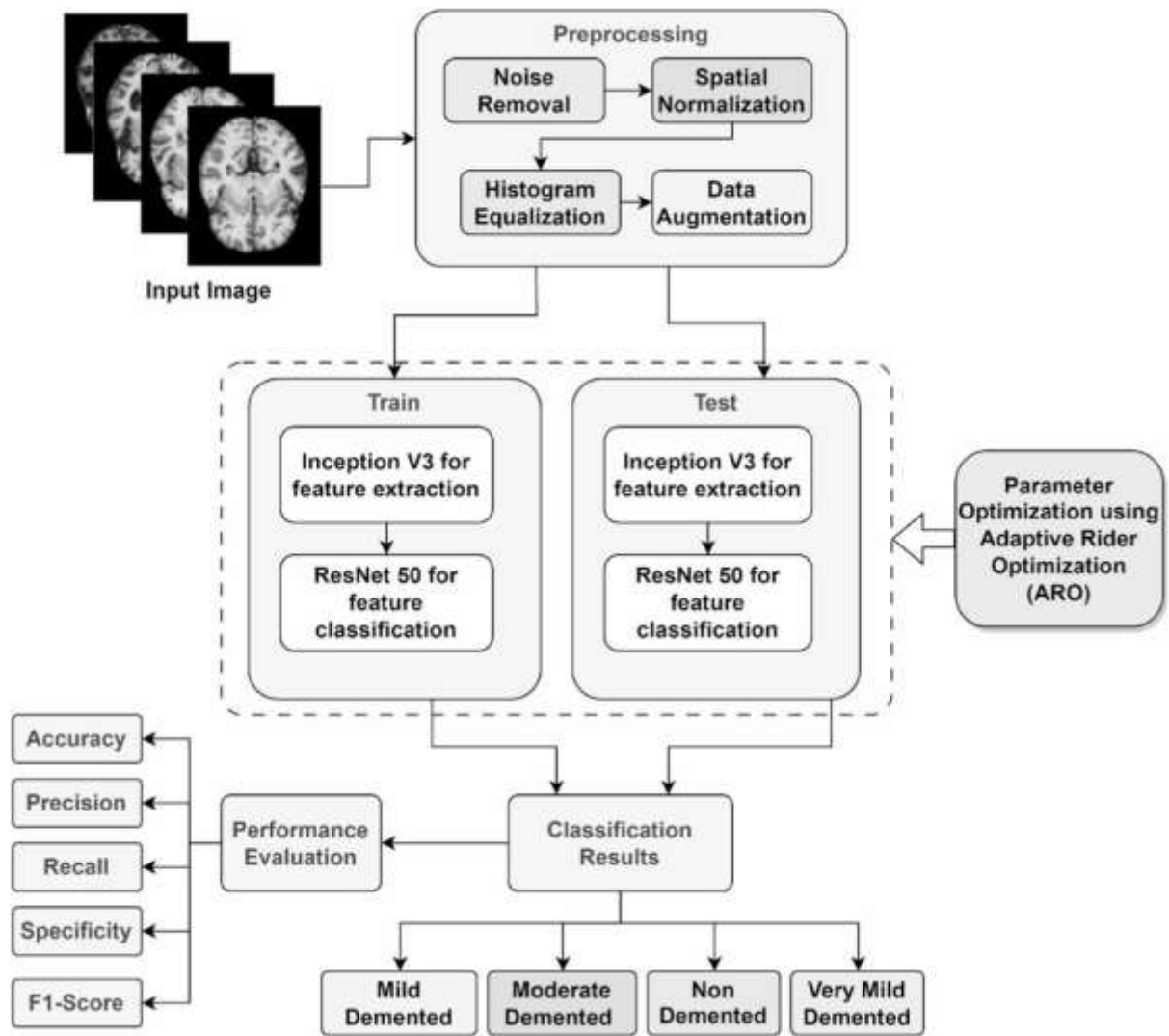
Article Number: 1664418

DOI: 10.3389/fnins.2025.1664418

Topic: Transfer Learning using ResNet50 for medical-image classification

The article was published on **15 December 2025** and investigates transfer learning using ResNet50 for Alzheimer's disease detection from MRI images.





4

Figure 1. Transfer learning using a pre-trained ResNet50 for medical-image classification.

2. Research Problem

Alzheimer's disease is a progressive neurological disorder that requires early and accurate diagnosis. Traditional diagnosis based on clinical examination and medical imaging can be difficult and time-consuming.

The research investigates whether **transfer learning with ResNet50** can improve Alzheimer's disease classification from MRI images.

The major challenge is that deep learning models generally require large amounts of labeled data. Medical datasets are often comparatively limited, making training a deep CNN from scratch difficult.

Limitations of existing approaches

Existing approaches may suffer from:

- Limited medical training data
- High computational requirements
- Overfitting

- Difficulty extracting discriminative MRI features
- Limited generalization

Transfer learning addresses some of these problems by reusing features learned from a large source dataset.

3. Objectives of the Study

The major objectives are:

1. Develop an Alzheimer's disease classification system using deep learning.
2. Apply **transfer learning using ResNet50**.
3. Extract meaningful features from MRI images.
4. Improve classification performance.
5. Investigate the effectiveness of pre-trained CNN features for medical-image diagnosis.
6. Provide a more automated approach to Alzheimer's detection.

4. Dataset Description

The study uses MRI data from the **Alzheimer's Disease Neuroimaging Initiative (ADNI)**. The ADNI dataset contains brain-imaging information used extensively for Alzheimer's disease research.

Type of data

The dataset consists of:

MRI brain images

Data processing

The images are prepared before being given to the neural network.

Typical preprocessing includes:

- Image resizing
- Normalization
- Data organization according to disease classes
- Preparation of training and testing data

The objective is to ensure that the images have a consistent format suitable for ResNet50.

5. Proposed Methodology

The overall methodology can be represented as:

MRI Images

↓

Preprocessing

↓

Image Resizing



Normalization



Pre-trained ResNet50



Feature Extraction



Fine-tuning



Classification Layer



Alzheimer's Prediction

The authors use **ResNet50** as the backbone.

Step 1 – Input

MRI brain images are provided to the system.

Step 2 – Preprocessing

The images are converted into a format suitable for the CNN.

Step 3 – Transfer Learning

A ResNet50 model pre-trained on a large image dataset is used.

Step 4 – Feature Extraction

The convolutional layers extract:

- Edges
- Textures
- Shapes
- Structural patterns

Step 5 – Fine-tuning

The model is adapted to the medical-image classification task.

Step 6 – Classification

The extracted features are passed through the classification layers to predict the Alzheimer's-related class.

6. Deep Learning Technique Used — Transfer Learning

What is Transfer Learning?

Transfer learning is a technique in which knowledge learned from one task is reused for another related task.

Instead of:

Random CNN



Train from scratch

we use:

Pre-trained CNN



Reuse learned features



Fine-tune for target problem

How ResNet50 is used

The pre-trained ResNet50 contains many convolutional layers that have already learned useful visual representations.

Initially:

Early layers → Frozen

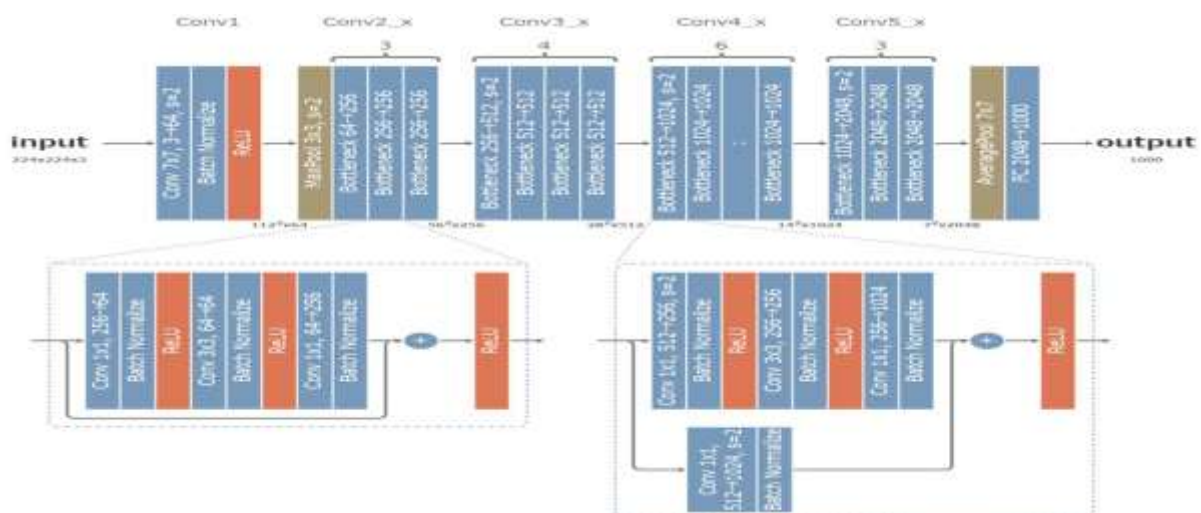
Middle layers → Frozen

Final layers → Fine-tuned

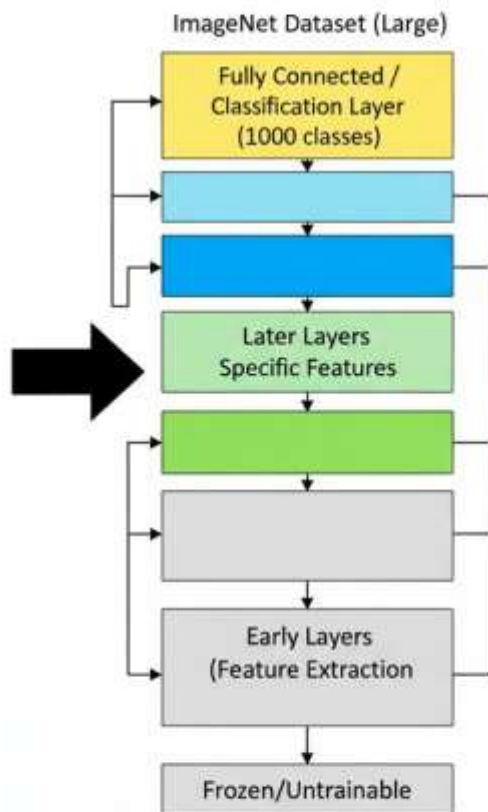
Classification layer → Replaced

The early layers learn general features, while deeper layers can be adapted to MRI-specific patterns.

7. Model Architecture



Pretained ResNeT-50 on ImageNet



Fine-tuning on Small New Dataset

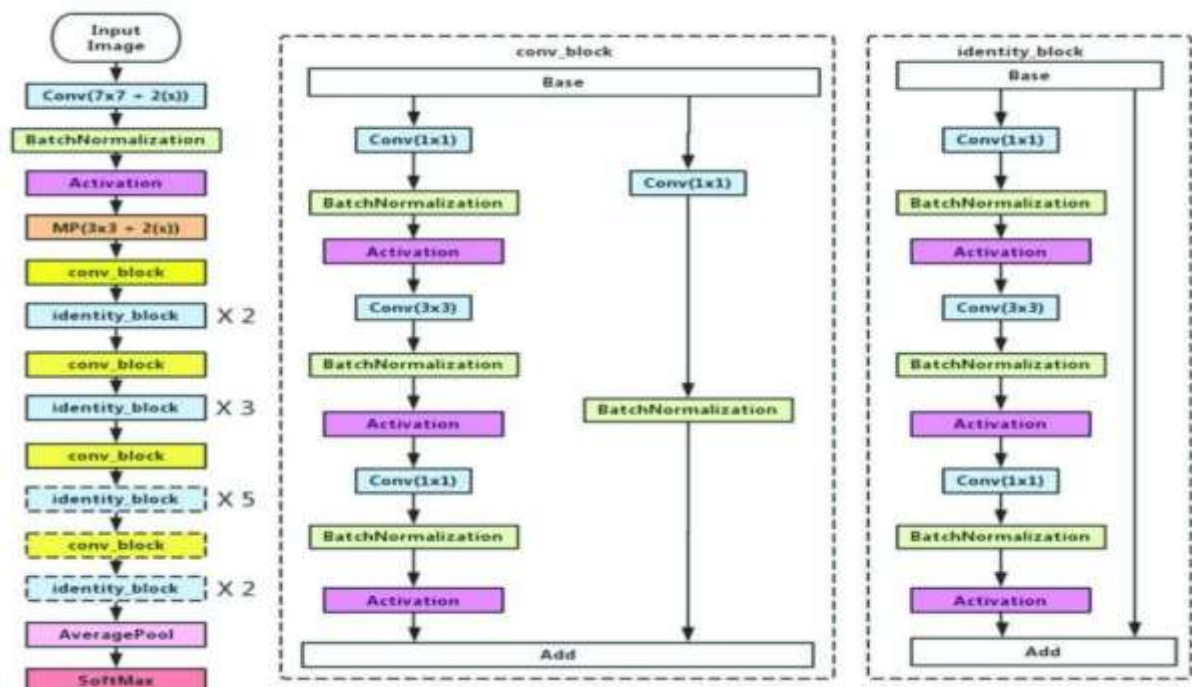
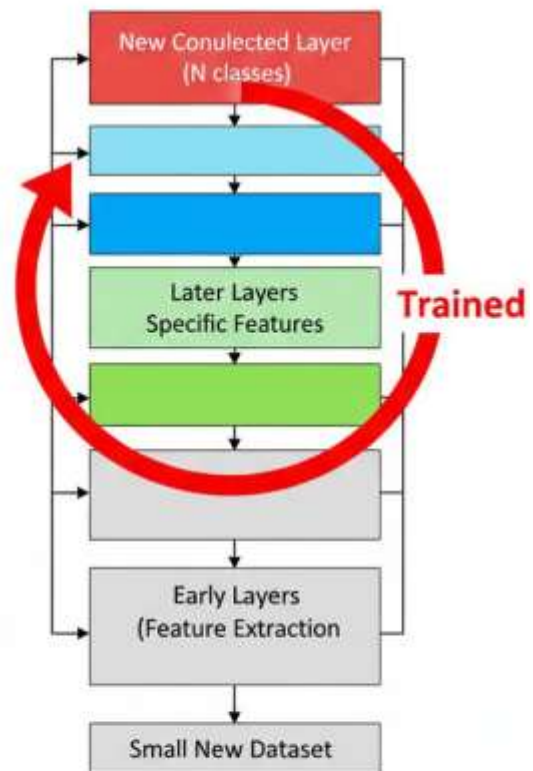
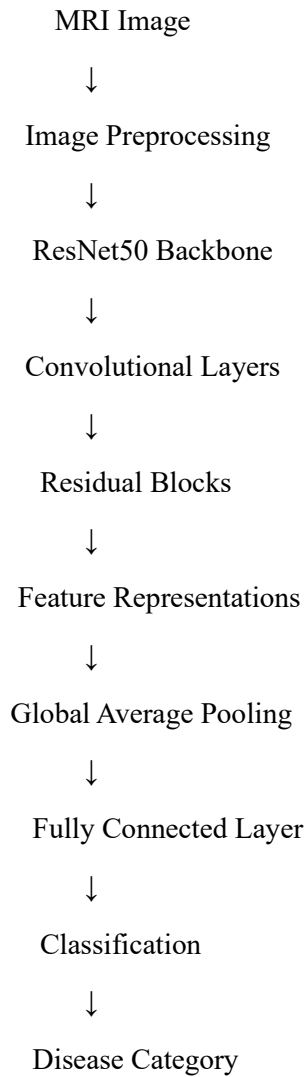


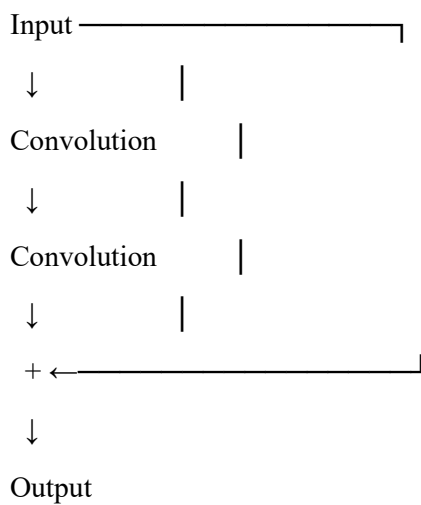
Figure 2. ResNet50 architecture showing residual blocks and skip connections.

Simplified architecture



Residual connection

One important feature of ResNet is the **skip connection**:



Skip connections help information and gradients flow through deep networks.

8. Performance Evaluation

The study evaluates the classification system using performance measures appropriate for medical-image classification.

Important metrics include:

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

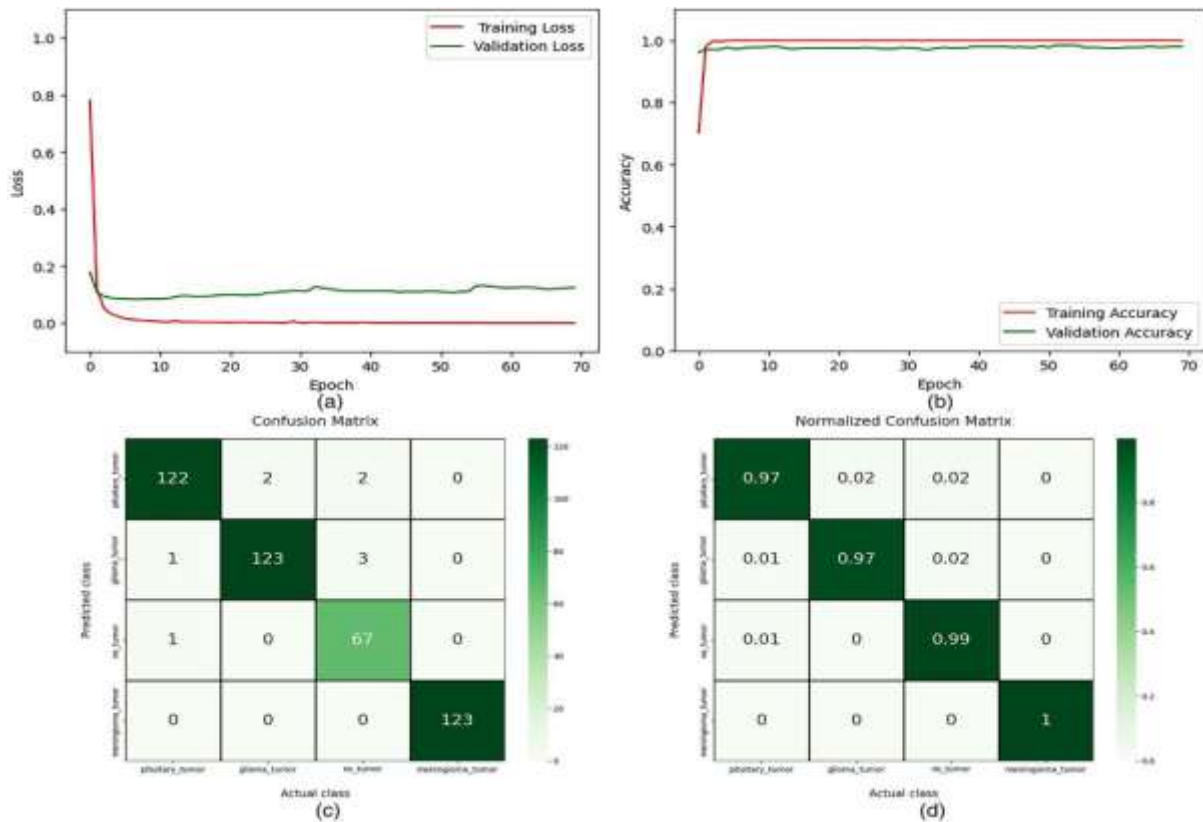
$$Recall = \frac{TP}{TP + FN}$$

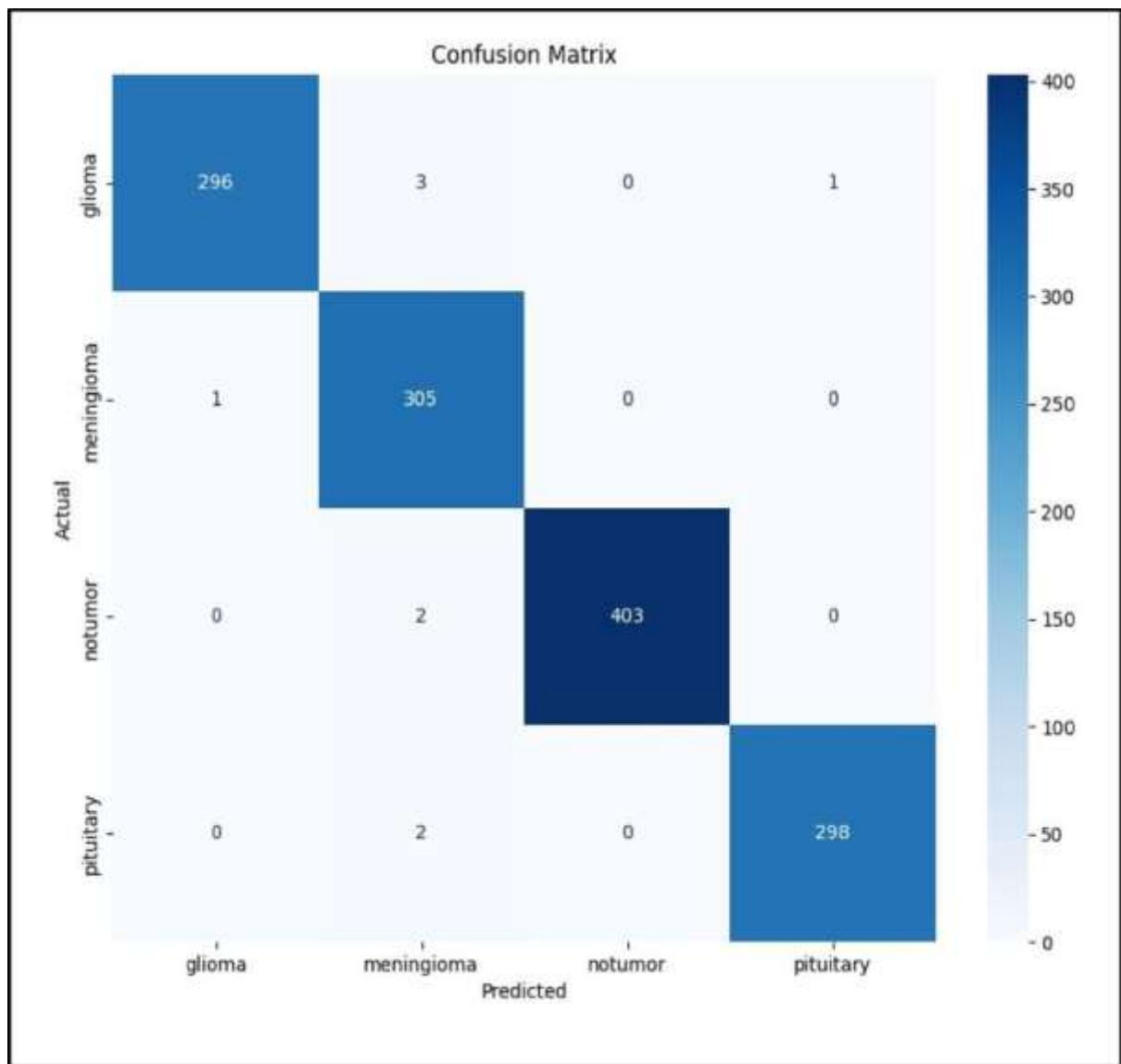
F1-score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

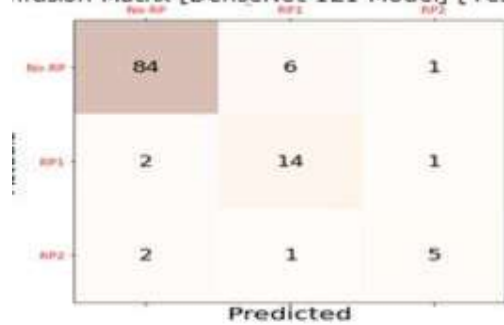
ROC-AUC

ROC-AUC can also be used to assess how well the model separates different classes.



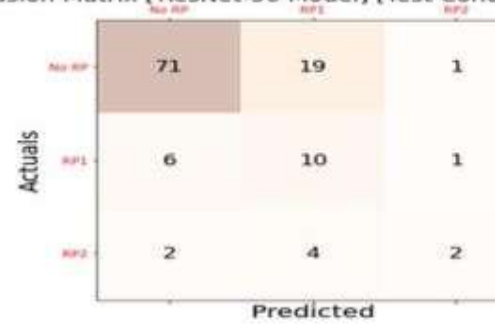


Confusion Matrix [DenseNet-121 Model] [Test Cohort]



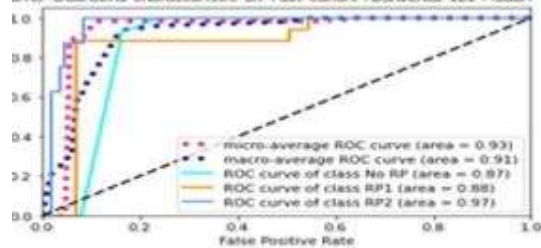
(a)

Confusion Matrix [ResNet-50 Model] [Test Cohort]



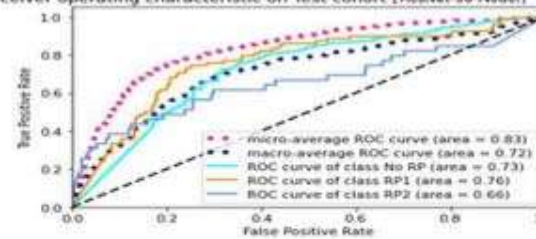
(b)

Receiver operating characteristic on Test cohort [DenseNet-121 Model]



(c)

Receiver operating characteristic on Test cohort [ResNet-50 Model]



(d)

Figure 3. Typical performance-evaluation visualization using confusion matrix and ROC curve.

Major result

The study demonstrates that **ResNet50-based transfer learning can effectively classify Alzheimer's-related MRI images**, supporting the use of transfer learning for medical-image diagnosis. The paper specifically focuses on transfer learning with ResNet50 and ADNI data.

9. Results and Discussion

The major findings are:

- Transfer learning is effective for medical-image classification.
- ResNet50 provides strong feature extraction.
- Pre-trained features reduce the need to train a deep CNN completely from scratch.
- Fine-tuning allows the network to adapt its deeper representations to the target medical task.
- The approach demonstrates potential for automated Alzheimer's detection.

Why does ResNet perform well?

The major reasons include:

1. Deep feature extraction
2. Residual/skip connections
3. Transfer of previously learned visual knowledge
4. Fine-tuning of task-specific layers
5. Ability to learn complex image patterns

10. Advantages of the Proposed Model

1. Reduced training requirement

Transfer learning reduces the need for a huge labeled medical dataset.

2. Better feature extraction

ResNet50 has already learned useful visual representations.

3. Reduced training time

The model does not need to learn all features from random initialization.

4. Deep architecture

ResNet50 can learn complex patterns from MRI images.

5. Good potential for medical diagnosis

The approach can support automated analysis and potentially assist healthcare professionals.

11. Limitations of the Study

Some limitations that should be considered are:

1. Dataset limitation

Medical datasets may not represent every hospital, population, scanner, or imaging protocol.

2. Generalization

A model trained on one dataset may not perform equally well on external datasets.

3. Computational requirements

ResNet50 is a relatively deep CNN and requires significant computational resources for training/fine-tuning.

4. Interpretability

Deep CNN predictions can be difficult to explain clinically.

5. Data imbalance

If some disease categories contain fewer samples, classification performance can be biased.

12. Critical Analysis

Is ResNet50 appropriate?

Yes.

ResNet50 is appropriate because the problem involves image classification and the available medical data can benefit from transfer learning.

Is the dataset adequate?

The ADNI dataset is highly relevant for Alzheimer's research, but **external validation** on independent clinical datasets is important before real-world deployment.

Are the metrics appropriate?

Yes.

Accuracy, precision, recall, F1-score and ROC-AUC provide a more complete evaluation than accuracy alone.

For medical diagnosis, **recall/sensitivity** is particularly important because missing a disease case can have serious consequences.

Are the results convincing?

The results support the usefulness of ResNet50 transfer learning, but stronger evidence would come from:

- Multiple independent datasets
- External validation
- Statistical significance testing
- Comparison with newer architectures

Could another technique perform better?

Potential alternatives include:

- DenseNet
- EfficientNet
- Vision Transformer
- ConvNeXt
- Hybrid CNN–Transformer models

DenseNet is particularly interesting because its dense connections promote feature reuse and improve gradient propagation. The original DenseNet study reported strong benchmark performance while using fewer parameters than many conventional architectures.

13. Future Scope

Future improvements could include:

1. Larger datasets

Use larger and more diverse MRI datasets.

2. External validation

Test the model on MRI images from different hospitals.

3. Explainable AI

Use techniques such as:

- Grad-CAM
- SHAP
- Saliency maps

to explain predictions.

4. Hybrid architectures

Combine ResNet with:

- Attention
- Transformers
- LSTM where appropriate

5. Better data augmentation

Use:

- Rotation
- Cropping
- Flipping
- Intensity variation

to improve generalization.

6. Medical-specific pre-training

Instead of using only generic ImageNet features, pre-train CNNs on large medical-image datasets.

Recent research has investigated medical-specific pretraining and found improvements over ImageNet initialization on several medical classification and segmentation tasks.

14. Conclusion

The reviewed research investigates **transfer learning using ResNet50 for Alzheimer's disease detection from MRI images**. The approach uses a pre-trained deep CNN to extract powerful image features and adapts the network to the medical classification task through fine-tuning.

The approach has important advantages, including reduced training requirements, strong feature extraction and improved efficiency compared with training a deep network entirely from scratch.

However, limitations such as dataset diversity, generalization, computational requirements and model interpretability must be addressed before clinical deployment.

Overall, **ResNet50-based transfer learning is an appropriate and promising approach for medical-image classification**, while future work should focus on larger datasets, external validation, explainability and medical-specific pretraining.

15. References

1. **Amine, J. M., & Mourad, M. (2025).** *Toward accurate Alzheimer's detection: transfer learning with ResNet50 for MRI-based diagnosis.* **Frontiers in Neuroscience, 19**, 1664418. DOI: 10.3389/fnins.2025.1664418.
2. **Huang, G., Liu, Z., van der Maaten, L., & Weinberger, K. Q. (2017).** *Densely Connected Convolutional Networks.* Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2261–2269. DOI: 10.1109/CVPR.2017.243.
3. **Hochreiter, S., & Schmidhuber, J. (1997).** *Long Short-Term Memory.* Neural Computation, 9(8), 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.